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GROUP 15

Module: Quantitative Methods For Finance

TOPIC: THE IMPACT OF NEWS SENTIMENT ON STOCK PRICES – AN ECONOMETRIC AND NLP-BASED ANALYSIS

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INTRODUCTION

Over the past two decades, the Vietnamese stock market has developed rapidly, with the VN30 Index emerging as the primary benchmark for blue-chip liquidity and capitalization on HOSE. As digital media accelerates information dissemination, investor decisions are increasingly shaped by fast-moving news.

Traditional theory, grounded in the Efficient Market Hypothesis (Fama, 1970), assumes that prices fully reflect available information. However, behavioral finance shows that biases and bounded rationality often drive markets - especially in emerging economies like Vietnam, where retail investors dominate trading. In such environments, sentiment not fully justified by fundamentals can move prices and generate anomalies.

This study combines NLP and panel econometrics to (i) classify Vietnamese financial news sentiment, (ii) measure the effects of firm-level and macroeconomic sentiment on VN30 stock returns, (iii) assess causality using fixed-effects and GMM models, (iv) test the persistence of sentiment effects, and (v) propose a reproducible sentiment–return framework for an emerging market context.

The key research questions are:

1. How does firm-specific sentiment influence subsequent VN30 returns?
2. How does macroeconomic sentiment affect individual stocks?
3. Are reactions asymmetric between positive and negative sentiment?
4. Does trading volume moderate these relationships?

Using a balanced panel of 11,010 firm-day observations from January 2023 to June 2024, the study captures a period of interest-rate fluctuations and post-pandemic recovery. Limitations include NLP challenges in Vietnamese and potential endogeneity in news coverage, partly addressed through dynamic panel models.

This research contributes new evidence on sentiment-driven price adjustments in Vietnam, offering implications for investors and policymakers.

LITERATURE REVIEW AND THEORETICAL BACKGROUND

I. Literature Review

Foundational work by Tetlock (2007) shows that pessimistic media tone predicts short-term market declines that later revert to fundamentals. Subsequent studies refined sentiment measurement through financial-specific dictionaries such as Loughran and McDonald (2011), while recent advances use transformer-based models like FinBERT (Araci, 2019), which significantly improves classification accuracy for financial texts.

Empirical findings consistently indicate that firm-level news sentiment drives immediate price adjustments, whereas macro sentiment influences broader market indices. In Vietnam, NLP-based analyses (e.g., Luu Hung et al., 2022) show that positive sentiment is associated with VN-Index gains, highlighting sentiment's predictive power beyond traditional fundamentals, particularly in panel settings

II. Theoretical Framework

1. Investor Sentiment and Noise Trading

Behavioral finance argues that prices may deviate from intrinsic values due to noise trading and cognitive biases. Sentiment - beliefs not fully supported by fundamentals - can materially influence trading behavior. Media coverage operates as a key transmission channel, where negative tone raises risk aversion and may depress prices temporarily, while positive tone can induce overreaction.

2. Hypotheses Development

H1: Firm-specific sentiment is positively associated with stock returns.
H2: Macroeconomic sentiment exerts a systematic effect on VN30 stock performance.
H3: Sentiment scores retain predictive power for future returns, controlling for market trends and trading volume.

DATA AND METHODOLOGY

I. Data Collection and Sample

The study utilizes a balanced panel dataset of firms listed in the VN30 Index on the Ho Chi Minh City Stock Exchange (HOSE).

The analysis spans 18 months, from January 1, 2023 to June 30, 2024. The data construction process involves merging two primary data sources:

Market Data: Daily trading variables - including closing prices, adjusted returns, trading volume, and VN30 benchmark returns - were obtained from Vietstock Finance.

Sentiment Data: Firm-specific and market-wide news articles were collected and processed using a Large Language Model (LLM). The model evaluated the textual content of each news item and generated a sentiment score from –1 (negative) to +1 (positive) for each entity on a daily basis.

II. Variable Definition

1. Dependent Variable

Daily Stock Return **($Ret_{i,t}$):** calculated as the logarithmic difference of the daily closing price:

$$Ret_{i,t} = \ln(P_{i,t}) - \ln(P_{i,t-1})$$

where $P_{i,t}$ is the closing price of firm *i* on day *t*.

2. Independent Variable

To investigate the predictive power of sentiment, we utilize the following lagged predictors:

Firm Sentiment ($SentFirm_{i,t-1}$): The aggregate sentiment score of news specifically related to firm *i* on day *(t-1)*. This variable captures investor psychology regarding specific corporate events.

Firm Sentiment Lag 2 ($SentFirm_{i,t-2}$): The firm sentiment score lagged by two periods *(t-2)*. This variable is employed as an instrumental variable in the GMM specification. It serves to identify the causal link between sentiment and returns by isolating variation in *t-1* sentiment that is uncorrelated with the error term at time *t*.

Macroeconomic Sentiment ($Sent\ macro_{t-1}$)

The macroeconomic sentiment variable captures the average daily sentiment score from market-wide news articles addressing inflation, monetary policy, GDP, and interest rates. Scores are extracted using a large language model and normalized to [−1, +1]. A one-day lag ensures macroeconomic information precedes price movements.

Lagged Market Return ($r_{m,t-1}$)

This variable represents the VN30 Index logarithmic return on day t−1, controlling for systematic market movements. The lagged specification avoids simultaneity bias and captures market-level shocks that spill over to firm-level returns.

Lagged Trading Volume ($Vol_{i,t-1}$)

Lagged trading volume is measured as the natural logarithm of firm i's trading volume on day t−1. The logarithmic transformation reduces right-skewness and stabilizes variance. This variable proxies for liquidity and investor attention.

Interaction Term ($Sent\ x\ Vol_{i,t-1}$)

The interaction term is the product of normalized firm-specific sentiment and normalized log trading volume. It tests whether sentiment impact varies with market attention. A positive coefficient indicates sentiment effects strengthen when trading activity is high, while a negative coefficient implies a dampening effect.

3. Econometric models

This section presents the econometric strategy employed to estimate the impact of sentiment indicators on stock returns. Given the structure of the dataset, which consists of repeated observations of multiple firms over time, panel data techniques are used to take advantage of both the cross-sectional and time-series dimensions. The modelling approach begins with a set of static panel estimators, followed by a dynamic specification to address potential endogeneity and return persistence.

3.1. Static Panel Models

a. Pooled Ordinary Least Squares (Pooled OLS)

The Pooled OLS estimator provides a baseline benchmark by treating the entire panel as a single pooled cross-section. The model assumes homogeneous intercepts and slopes across all firms:

$$ret_{i,t} = \beta_0 + \beta X_{i,t-1} + u_{i,t}$$

Where:

- $ret_{i,t}$: stock return of firm i at time t
- $X_{i,t-1}$: vector of all lagged explanatory variables
- β_0: common intercept
- β: vector of slope coefficients
- $u_{i,t}$: idiosyncratic error term

This specification does not account for firm-level heterogeneity. As such, it may produce biased estimates if unobserved firm-specific characteristics such as long-term risk exposure, industry affiliation, or governance structure are correlated with the explanatory variables. Nevertheless, Pooled OLS is useful as a benchmark for comparing more flexible estimators.

b. Fixed Effects Model (FEM)

The Fixed Effects estimator addresses unobserved heterogeneity by allowing each firm to have its own intercept:

$$ret_{i,t} = \alpha_i + \beta X_{i,t-1} + u_{i,t}$$

Where: α_i: firm-specific intercept capturing all time-invariant characteristics

Here, α_i captures all firm-specific, time-invariant characteristics. FEM is particularly appropriate when these characteristics are correlated with the sentiment variables for instance, when firms differ systematically in market risk, investor base, or information transparency. By eliminating time-invariant heterogeneity, FEM provides consistent estimates even when the unobserved fixed effects are correlated with regressors.

c. Random Effects Model (REM)

The Random Effects estimator instead models unobserved heterogeneity as a random disturbance:

$$ret_{i,t} = \beta_0 + \beta X_{i,t-1} + \mu_i + u_{i,t}$$

where μ_i is the firm-specific random effect. REM relies on the crucial assumption that:

$$Cov\ (X_{i,t-1,\mu_i}) = 0$$

Under this assumption, REM is more efficient than FEM due to its use of both within-firm and between-firm variation. REM also permits estimation of coefficients on time-invariant variables, which FEM inherently absorbs.

If the Hausman specification test fails to reject the null hypothesis of orthogonality, REM is the preferred estimator because it remains consistent and yields smaller standard errors.

d. Baseline Panel Regression Equation

All static models are estimated using the following baseline specification:

$$ret_{i,t} = \alpha + \beta_1 sent_{firm,i,t-1} + \beta_2\ sent_{firm,i,t-2} + \beta_3 sent_{macro,t-1}$$

$$+ \beta_4 market\ ret_{t-1} + \beta_5\ log\ volume_{i,t-1} + \beta_6\ (sent\ x\ vol)_{i,t-1} + u_{i,t}$$

This model incorporates both sentiment indicators and control variables that capture market-wide movements and liquidity effects. The interaction term allows the analysis to assess whether trading volume amplifies or attenuates the effect of sentiment on stock returns.

3.2. Dynamic Panel Model (Robustness Check)

System GMM (Arellano–Bond)

To address potential endogeneity and dynamic return behavior, a dynamic panel specification is estimated using the Arellano–Bond Generalized Method of Moments (GMM). Stock returns often display short-run persistence, and incorporating the lagged dependent variable introduces bias in static estimators such as FEM and REM:

$$ret_{i,t} = p\ ret_{i.t-1} + \beta X_{i,t-1} + u_{i,t}$$

Because $ret_{i,t-1}$ is endogenous, System GMM uses deeper lags of the explanatory variables as internal instruments to obtain consistent estimates. This approach simultaneously corrects for: autocorrelation in returns, endogeneity in sentiment variables, unobserved firm-level heterogeneity, heteroskedasticity and measurement error.

The dynamic estimator thus serves as an essential robustness check, ensuring that the empirical results are not driven by misspecification or omitted dynamics in the return-generating process.

III. Model Diagnostics and Specification

This section outlines the diagnostic procedures and econometric specification tests conducted to ensure the validity, consistency, and robustness of the regression models employed in this study. Because panel data may violate classical regression assumptions, the following diagnostics are essential for determining the appropriate model structure Pooled OLS, Fixed Effects (FEM), Random Effects (REM), or the dynamic GMM estimator and for verifying whether the econometric assumptions underlying each model are satisfied.

1. Hypothesis Testing Model

The empirical framework is designed to test whether sentiment-related variables and trading activity have a statistically meaningful influence on stock returns within the VN30 universe. The hypotheses associated with the key regression parameters are summarized below:

Hypothesis 1: Effect of Firm-Specific Sentiment

H1: $\beta_1 > 0$

This tests whether positive sentiment at the firm level leads to higher subsequent stock returns. It reflects the idea that investors may react to favorable news by purchasing shares, thereby driving prices upward.

Hypothesis 2: Neutral Role of Macro Sentiment

H2: $\beta_3 = 0$

This hypothesis asserts that macroeconomic sentiment does not exert an additional influence on individual stock returns once systematic market movements - captured through lagged market returns - are controlled for. This aligns with the notion that macro news largely affects market-wide behavior rather than firm-level return variations.

Hypothesis 3: Moderating Role of Trading Volume

H3: $\beta_6 \neq 0$

The interaction term between sentiment and trading volume tests whether investor attention strengthens or dampens the relationship between sentiment and future returns. A significant coefficient would support behavioral finance theories suggesting that sentiment-driven mispricing is amplified during periods of heightened trading activity.

2. Diagnostic Tests

To identify the appropriate panel-data model and ensure that the estimated coefficients are statistically reliable, several diagnostic tests are conducted. These tests address common econometric concerns including model selection, multicollinearity, heteroskedasticity, autocorrelation, and instrument validity in dynamic panels.

2.1. Variance Inflation Factor (VIF) - Test for Multicollinearity

Purpose:
To assess whether independent variables are excessively correlated, which may distort coefficient estimates and inflate standard errors.

Hypotheses:

H_0: No multicollinearity among regressors

H_A: Multicollinearity is present

If the result of the VIF test is:

Acceptable	VIF = 1: No multicollinearity	1 < VIF < 5: Mild multicollinearity
Unacceptable	VIF > 5: Strong multicollinearity	VIF > 10: Severe multicollinearity

Reject H0 if VIF > 5 as the model is exposed to strong multicollinearity.

2.2. Breusch - Pagan Lagrange Multiplier (LM) Test - Pooled OLS vs. Random Effects

Purpose:
To determine whether panel-level variance components exist. If significant, Pooled OLS becomes inappropriate because it assumes no heterogeneity across firms.

Hypotheses:

H_0: No panel effects → Pooled OLS is appropriate

H_A: Panel effects exist → REM or FEM should be used

Decision Rule:

p-value > 0.05: Fail to reject H_0 → Use Pooled OLS

p-value < 0.05: Reject H_0 → Use Random Effects or Fixed Effects

This test serves as the first step before performing the Hausman test.

2.3. Hausman Specification Test - Fixed Effects vs. Random Effects

Purpose:
To determine whether the unobserved firm-specific components (u_i) are correlated with the regressors.

Hypotheses:

H_0: Random Effects model is consistent and efficient

H_A: Fixed Effects model is consistent (correlation exists)

Decision Rule:

p-value > 0.05: Fail to reject H_0 → Random Effects preferred

p-value < 0.05: Reject H_0 → Fixed Effects preferred

Interpretation:
If REM is chosen, the model assumes that firm-specific unobserved heterogeneity does not bias coefficient estimates and allows for efficiency gains.

2.4. Tests for Autocorrelation

Autocorrelation violates classical regression assumptions and can bias coefficient standard errors. Two types of autocorrelation tests are applied:

Wooldridge Test for Autocorrelation in Panel Data

- Widely used for static models.

- Detects first-order serial correlation.

Durbin - Watson Test

- Additional check for serial dependence in residuals.

Hypotheses:

H_0: No first-order autocorrelation

H_A: First-order autocorrelation is present

Interpretation:

p-value > 0.05: No serial correlation

p-value < 0.05: Serial correlation exists → cluster-robust standard errors required

2.5. Heteroskedasticity Tests

Panel data often exhibit non-constant variance in error terms across firms.

Breusch - Pagan Test / Modified Wald Test

Hypotheses:

H_0: Homoskedasticity (constant error variance)

H_A: Heteroskedasticity exists

Interpretation:

p-value < 0.05: Evidence of heteroskedasticity → use robust or clustered SE

p-value > 0.05: Variance is constant across firms

Robust estimation methods (e.g., "clustered" covariance estimators) ensure valid statistical inference even when heteroskedasticity is present.

2.6. Dynamic Model Diagnostics (If Using GMM)

Because the Arellano - Bond GMM estimator is used as a robustness check, additional tests must verify the validity of instruments and serial correlation structures.

AR(1) and AR(2) Tests (Arellano–Bond)

AR(1) is expected to be significant because differencing induces first-order correlation.

AR(2) must be insignificant for the model to be valid.

Decision Rule:

AR(1) p-value < 0.05 → expected

AR(2) p-value > 0.05 → no second-order autocorrelation → instruments valid

Hansen J-Test for Overidentifying Restrictions

Purpose:
To test whether the instruments used in GMM are valid (uncorrelated with the error term).

Hypotheses:

H_0: Instruments are valid

H_A: Instruments are invalid

Decision:

p-value > 0.10: Fail to reject H_0 → Instruments valid

p-value < 0.05: Reject H_A → Model suffers from instrument invalidity

Note:
Extremely large p-values (close to 1) may indicate instrument proliferation.

DATA ANALYSIS AND INTERPRETATION

I. Descriptive Statistics

	Log_Return	Company_sentiment_score	Market_Return	Macro_Sentiment
count	11010.000000	11010.000000	11010.000000	11010.000000
mean	0.000503	0.039509	0.000474	0.116987
std	0.017235	0.251363	0.010174	0.189812
min	-0.072571	-1.000000	-0.048132	-0.820000
25%	-0.008176	0.000000	-0.004113	0.010000
50%	0.000000	0.000000	0.001516	0.115000
75%	0.008668	0.000000	0.006206	0.232000
max	0.067630	1.000000	0.033864	0.630000

Figure 1. Statistical Results

Source: Research Group

Table 1 reports the descriptive statistics of the variables. Stock returns (Log_Return) display a mean near zero (0.05%) and moderate variability (standard deviation of 0.0172) which is consistent with the typical behavior of return series. The wide range of returns (from –7.26% to 6.76%) indicates occasional large price movements, which is typical in emerging equity markets.. The median of zero reflects the prevalence of trading days with minimal price changes.

Lagged firm-level sentiment (Company_Sentiment) exhibits a slightly positive mean (approximately 0.0395) but substantial dispersion (standard deviation of 0.2514). Its bounded range from –1 to 1 reflects the normalized nature of sentiment scores derived from news headlines. The relatively high standard deviation compared to the mean suggests news sentiment varies considerably from day to day and firm to firm.

Macro-level sentiment (Macro_Sentiment) shows a higher mean (0.117) compared to firm-specific news. Its standard deviation (0.1898) is lower than that of firm sentiment, indicating that aggregated macroeconomic news tends to fluctuate less than idiosyncratic firm news, reflecting its aggregated nature.

Lagged market returns (Market_Return) exhibit a mean close to zero (0.05%) and a relatively small standard deviation (0.0102). This lower volatility, compared to individual

stock returns, illustrates the diversification effect inherent in the broad market index (VN30).

Overall, the statistics indicate substantial heterogeneity in sentiment and trading activity, while return measures exhibit patterns consistent with established financial data characteristics in emerging markets.

II. Correlation matrix

	Log_Return	**Company_sentiment_score**	**Market_Return**	**Macro_Sentiment**	**Log_Volume**	**Log_return_lag1**
Log_Return	1.0000	0.0581	0.6077	0.0102	0.0582	-0.0118
Company_sentiment_score	0.0581	1.0000	0.0373	0.0107	0.0257	0.0149
Market_Return	0.6077	0.0373	1.0000	-0.0005	-0.0319	-0.0077
Macro_Sentiment	0.0102	0.0107	-0.0005	1.0000	0.0345	-0.0523
Log_volume	0.0582	0.0257	-0.0319	0.0345	1.0000	0.0473
Log_return_lag1	-0.0118	0.0149	-0.0077	-0.0523	0.0473	1.0000

Figure 2. Correlation Matrix

Source: Research Group

The correlation matrix indicates that stock returns (Log_Return) are only weakly related to the other variables. The strongest association is a high correlation of 0.6077 with market returns (Market_Return), while correlations with sentiment measures remain negligible (0.0581 with firm sentiment and 0.0102 with macro sentiment). Firm-level sentiment (Company_sentiment_score) shows very weak correlations across the board, with the highest being 0.0373 with market returns, suggesting that firm-specific news is largely idiosyncratic. Macro sentiment (Macro_Sentiment) behaves similarly, displaying near-zero correlations with all variables, including market returns (–0.0005). Trading activity, measured by log volume (Log_Volume), also exhibits minimal associations (e.g., 0.0582 with stock returns and 0.0345 with macro sentiment). Lagged returns (Log_return_lag1) show virtually no correlation with contemporaneous returns (–0.0118) or other predictors. Overall, the pairwise correlations are low, suggesting no immediate concern of multicollinearity. This is further validated by the VIF results reported in the diagnostics section, confirming that the explanatory variables do not exhibit harmful linear dependence.

III. Model Diagnostics and Specification

1. Model Comparison

Table 3 presents a comparison of three panel regression specifications - Pooled OLS, Fixed Effects (FEM), and Random Effects (REM) - estimating the impact of news sentiment, market returns, and trading activity on stock returns. All models use 11,010 observations and report standard errors to ensure the robustness of the estimates. The coefficients are broadly stable across models. Market returns (Market_Return) show a consistently positive and highly significant effect across all specifications (coefficients = 1.03), indicating that systematic risk plays a dominant role in driving individual stock performance.

Crucially, Firm-level sentiment (Company_sentiment_score) exhibits a consistently positive and statistically significant association with returns across both Fixed Effects (β= 0.0021) and Random Effects (β = 0.0023) specifications. Trading volume (Log_Volume) maintains a significant positive effect, supporting the volume-price confirmation hypothesis. Conversely, Macro-level sentiment (Macro_Sentiment) and Lagged Returns do not show statistically meaningful effects, with small coefficients and p-values exceeding conventional significance levels. Overall, the results suggest that while market-wide factors play a substantial role, firm-specific sentiment measures provide additional, statistically significant explanatory power for short-term return dynamics.

Model Comparison

	(1) Pooled OLS	(2) Random Effects	(3) Fixed Effects
Dep. Variable	Log_Return	Log_Return	Log_Return
Estimator	PooledOLS	RandomEffects	PanelOLS
No. Observations	11010	11010	11010
Cov. Est.	Clustered	Clustered	Clustered
R-squared	0.3766	0.3766	0.3844
R-Squared (Within)	0.3788	0.3788	0.3844
R-Squared (Between)	-0.2793	-0.2793	-9.0116
R-Squared (Overall)	0.3766	0.3766	0.3523
F-statistic	1329.3	1329.3	1370.6
P-value (F-stat)	0.0000	0.0000	0.0000
const	-0.0144	-0.0144	-0.0431
	(-7.3335)	(-7.3335)	(-6.9866)
Company_sentiment_score	0.0023	0.0023	0.0021
	(3.2598)	(3.2598)	(2.9537)
Market_Return	1.0313	1.0313	1.0398
	(17.721)	(17.721)	(17.561)
Macro_Sentiment	0.0006	0.0006	0.0001
	(1.1893)	(1.1893)	(0.2243)
log_volume	0.0010	0.0010	0.0029
	(7.1154)	(7.1154)	(6.9931)
log_return_lag1	-0.0109	-0.0109	-0.0165
	(-0.6790)	(-0.6790)	(-1.0138)

T-stats reported in parentheses

Figure 3. Model Comparison

Source: Research Group

2. Diagnostic Tests & Model Selection

First, to determine the choice between Fixed Effects (FEM) and Random Effects (REM), the Hausman specification test was performed. The test yields a Chi-square statistic of **1.8308** with a corresponding P-value of **0.8720**.The results indicate that the sign and statistical significance of the key variable of interest (Sentiment) are consistent across both models, confirming the robustness of the findings. Since the P-value is significantly greater than the conventional 0.05 threshold, the test failed to reject the null hypothesis. Under the null hypothesis of the Hausman test, the REM model yields more efficient estimates. Therefore, Random Effects is the most appropriate model.

Second, to confirm the validity of using a panel data approach over a simple Pooled OLS

regression, the Breusch-Pagan LM test was conducted. The results show a Lagrange Multiplier (LM) statistic of **298.89** with a P-value effectively equal to 0.0000 (reported as **1.73e-62**). This result strongly rejects the null hypothesis, confirming the presence of panel effects across the VN30 firms. Consequently, the simple Pooled OLS model is unsuitable and the Random Effects Model (REM) is preferred.

Concurrently, the results show that Variance Inflation Factor (VIF) values for all independent variables are exceptionally low, hovering around 1.00, < 2. Specifically, the VIFs for Company_sentiment_score (1.002), Market Return (1.002), and Macro_Sentiment (1.004) are far below the critical threshold of 5 or 10. This guaranteed that the regression coefficients are estimated with precision and stability. Overall, the diagnostics support the use of the REM specification. To ensure the robustness of the estimates, we examined the model's residual properties. The Durbin-Watson test yields a statistic of **d = 1.9904**, which is asymptotically close to 2.0, indicating no first-order autocorrelation. This confirms that the error terms are not serially correlated

```
>>> HAUSMAN TEST
Chi-Square: 1.8308
P-value: 0.8720

>>> BREUSCH-PAGAN TEST
LM: 298.8929997896667 p-value: 1.732422710627136e-62

>>> VIF
                Feature        VIF
0                 const  116.047826
1  Company_sentiment_score    1.002425
2           Market_Return    1.002520
3         Macro_Sentiment    1.004232
4             log_volume    1.005289
5         log_return_lag1    1.005423

>>> DURBIN-WATSON
1.9904438859077878
```

Figure 4. Model diagnostic test results

Source: Research Group

IV. Empirical Results

1. Random Effects Results

```
=== 6.3 Panel Random Effects Results ===
                        RandomEffects Estimation Summary
================================================================================
Dep. Variable:              Log_Return   R-squared:                       0.3766
Estimator:               RandomEffects   R-squared (Between):            -0.2793
No. Observations:                11010   R-squared (Within):              0.3788
Date:                 Tue, Dec 02 2025   R-squared (Overall):             0.3766
Time:                         03:22:58   Log-likelihood                3.169e+04
Cov. Estimator:              Clustered
                                         F-statistic:                     1329.3
Entities:                           30   P-value                          0.0000
Avg Obs:                        367.00   Distribution:                F(5,11004)
Min Obs:                        367.00
Max Obs:                        367.00   F-statistic (robust):            70.589
                                         P-value                          0.0000
Time periods:                      367   Distribution:                F(5,11004)
Avg Obs:                        30.000
Min Obs:                        30.000
Max Obs:                        30.000

                                     Parameter Estimates
==========================================================================================
                         Parameter  Std. Err.     T-stat    P-value    Lower CI    Upper CI
------------------------------------------------------------------------------------------
const                      -0.0144     0.0020    -7.3335     0.0000     -0.0182     -0.0105
Company_sentiment_score     0.0023     0.0007     3.2598     0.0011      0.0009      0.0037
Market_Return               1.0313     0.0582     17.721     0.0000      0.9172      1.1454
Macro_Sentiment             0.0006     0.0005     1.1893     0.2343     -0.0004      0.0017
log_volume                  0.0010     0.0001     7.1154     0.0000      0.0007      0.0012
log_return_lag1            -0.0109     0.0160    -0.6790     0.4972     -0.0423      0.0206
==========================================================================================
```

Figure 5. Random Effects Results

Source: Research Group

The estimation results derived from the Random Effects (REM) model provide a comprehensive view of the determinants of stock returns in the VN30 index. The model demonstrates robust explanatory power with an overall R-squared of 0.3766, indicating that nearly 38% of the return variation is explained by the selected covariates. Furthermore, the highly significant F-statistic of 1329.3 ($p < 0.001$) confirms the joint validity of the regressors. The constant term is negative and statistically significant ($\beta = -0.0144$), representing the baseline return when all explanatory variables are zero. To ensure the reliability of these inferences, particularly given the presence of heteroskedasticity detected

in preliminary tests, all coefficients are reported with clustered standard errors. Firm-level sentiment exhibits a positive and statistically significant coefficient (β = 0.0023, t = 3.26, $p < 0.01$) in the panel specifications (FEM and REM). This finding indicates that firm-specific sentiment has a direct and immediate positive impact on stock returns. Specifically, a one-unit increase in the sentiment score is associated with a 0.23 percentage points increase in returns, holding other factors constant, a result that aligns with behavioral finance theories where investor sentiment drives temporary demand shifts. Unlike the insignificant results often found in static pooled OLS models in other studies, the significance here suggests that idiosyncratic information is a critical driver of price movements. The positive sign implies that the market reacts swiftly to "good news" specific to the enterprise, and this reaction is not subsumed by broader market movements.

Regarding the control variables, Market Return emerges as the most dominant determinant of individual stock performance. The coefficient is highly significant and close to unity (β = 1.0313, t = 17.72, $p < 0.001$). This strong positive relationship confirms that the sampled stocks exhibit a high sensitivity to systematic risk, moving almost 1-to-1 with the general market trend. Additionally, Trading Volume displays a consistent positive relationship with returns (β = 0.001, t = 7.12, $p < 0.001$). This outcome supports the liquidity premium theory, or more specifically in this short-run context, suggests that high-volume trading days are associated with positive price pressure.

In contrast to firm sentiment, the Macro_Sentiment variable is not statistically significant in this model (P-value = 0.2343 > 0.05), although the coefficient is positive (β = 0.0006). This may stem from the fact that the impact of macroeconomic information often has a longer lag and affects the entire market rather than individual stocks. Furthermore, much of the impact of macro news might have already been captured by the Market Return variable, resulting in this variable no longer holding significant independent explanatory power in the firm-level panel data model.

2. Impacts On Sentiment Vs. Market Returns

The comparative analysis between market factors and sentiment variables reveals a clear hierarchy in the drivers of stock returns within the VN30. Consistent with standard asset pricing theories regarding systematic risk, the Market Return remains the dominant determinant of individual stock performance. With a coefficient of approximately 1.03 (t = 17.72), the results indicate that VN30 stocks are highly sensitive to general market movements, suggesting that broad trends dictate the primary direction of price changes.

However, despite the overwhelming dominance of market returns, firm-specific sentiment demonstrates a distinct and statistically significant impact. The Company Sentiment Score yields a positive coefficient of 0.0023 (t = 3.2598). While the magnitude of this effect is considerably smaller than that of the market return, its statistical significance confirms that news sentiment provides incremental explanatory power. This finding implies that while investors primarily follow market trends, they still react efficiently to idiosyncratic news. Consequently, firm-level sentiment serves as a valid, independent signal for predicting short-term return deviations that cannot be explained by systematic risk alone.

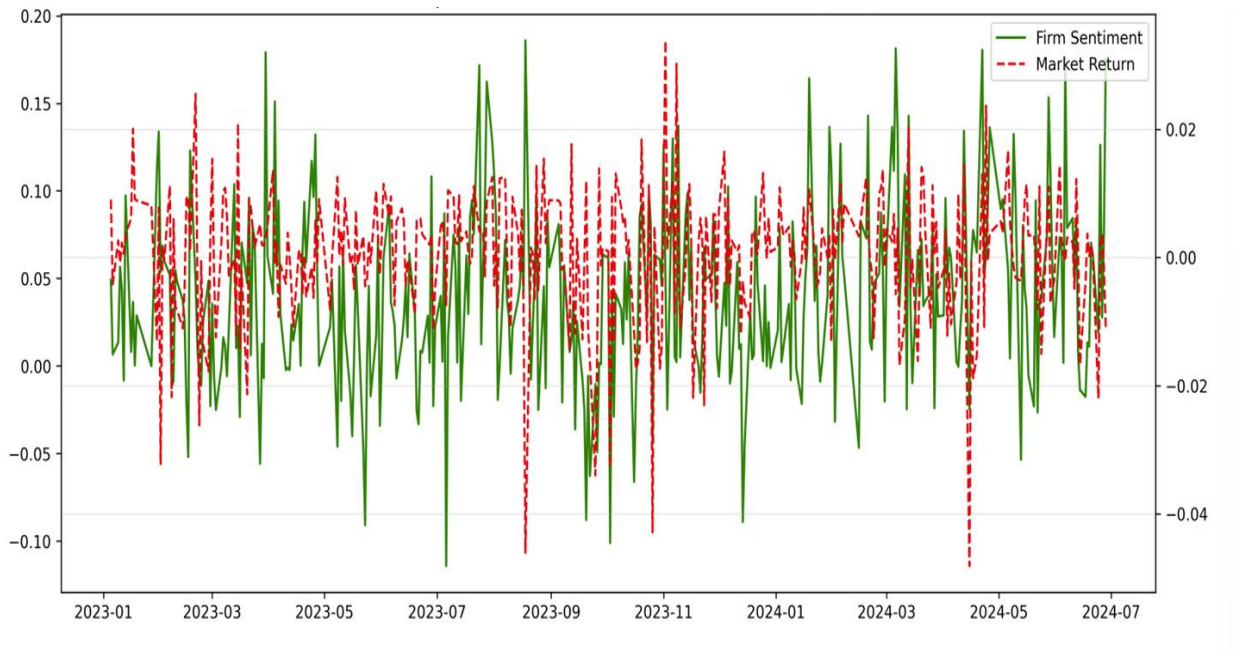

Figure 6. Relationship Between Firm Sentiment & Market Return (VN30)

Source: Research Group

CONCLUSION AND IMPLICATIONS

I. Conclusion

This study investigates the impact of news sentiment on stock returns within the VN30 index, yielding critical insights into the behavior of emerging markets. The empirical results from the Random Effects model confirm that firm-specific sentiment, extracted via Large Language Models, exerts a statistically significant positive influence on returns. This validates the hypothesis that linguistic data holds intrinsic predictive value beyond traditional financial metrics.

However, consistent with the Capital Asset Pricing Model, market risk remains the dominant driver of performance, indicating that systematic trends heavily dictate price movements. Furthermore, while trading volume positively affects returns, macroeconomic sentiment shows no significant direct impact in the short term. Ultimately, these findings support the Adaptive Market Hypothesis, suggesting that the Vietnamese market efficiently integrates qualitative information, allowing sentiment to serve as a valuable incremental signal alongside market fundamentals.

II. Implications

This study offers significant methodological and practical implications for financial analysis in emerging markets. Methodologically, it validates the efficacy of Large Language Models, specifically Claude, for extracting financial sentiment in the Vietnamese context, demonstrating that AI-driven approaches can capture nuances superior to traditional dictionary-based methods. Additionally, the research highlights the critical role of rigorous model diagnostics, as the Hausman test results confirmed the suitability of the Random Effects framework over Fixed Effects for the VN30 dataset, suggesting that specific firm-invariant characteristics are not the primary source of bias.

Practically, the findings justify the integration of sentiment analysis as a robust alternative data source for investors. The distinct positive correlation between firm-specific sentiment and returns suggests that linguistic signals can enhance short-term trading strategies by identifying mispricing not captured by market fundamentals. For corporate managers, the results underscore the financial materiality of media relations; maintaining a positive public image is not merely a branding exercise but a direct driver of stock valuation, necessitating proactive communication strategies.

III. Limitations and Future Directions

While this study provides robust evidence for the VN30 index, several limitations suggest promising avenues for future research. First, the reliance on daily data frequency may obscure the immediate, short-lived market reactions to news. Future studies could employ intraday or minute-level data to capture high-frequency volatility and determine the precise speed at which linguistic information is incorporated into prices.

Second, this research treats sentiment as a linear continuum; however, market psychology often exhibits asymmetry. Subsequent investigations should distinguish between positive and negative shocks to test if "fear" drives price corrections more potently than "greed." Third, the current pooled approach masks potential sectoral heterogeneity. Analyzing specific industries, such as Banking versus Real Estate, could reveal varying sensitivities to sentiment. Finally, exploring market microstructure, specifically the interaction between sentiment and order flow imbalance, would deepen the understanding of the liquidity mechanisms underlying these price adjustments.

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